Webserv Configuration Concepts

Configuration

A **configuration** is the collection of settings that define how a program should behave. Instead of modifying the source code, the program reads these settings at startup and adjusts its behavior accordingly.

Config

Config is simply the abbreviated form of **configuration**. Both words mean the same thing.

Directive

A **directive** is a single instruction in a configuration file. It consists of a keyword and one or more values that define one specific behavior of the program.

Example:

```
listen 8080;
root /var/www;
```

Each line is one directive.

Server Block

A **server block** is a group of related directives that describe the configuration of one virtual server (one website). It contains all the general rules that apply to that website, such as the listening port, domain name, document root, error pages, and locations.

Location Block

A **location block** is a subgroup inside a server block that defines how requests matching a specific URL path should be handled. It allows different parts of the same website to behave differently.

Example:

- `/images`
- `/uploads`
- `/cgi-bin`

Each location may override some of the server's general settings.

Website

A **website** is a collection of web resources (HTML pages, CSS, JavaScript, images, videos, APIs, etc.) that are accessible under a single domain name or virtual host.

Examples:

- `google.com`
- `youtube.com`
- `example.org`

In webserv, each website is typically represented by one server block.

Routing

Routing is the process of deciding which configuration or handler should process an incoming request based on information such as the requested URL.

In webserv, routing mainly means: 1. Selecting the correct server block (using the port and `Host` header). 2. Selecting the correct location block (using the request path).

Common Interview / Oral Questions

Why do we use a configuration file?

To separate the program's behavior from its source code, allowing users to change settings without recompiling the application.

Why not hard-code everything?

Hard-coding would require modifying, recompiling, and redeploying the program whenever a setting changes. Configuration makes the program reusable and flexible.

Why is the configuration hierarchical?

Because settings naturally have different scopes:

- Global behavior belongs to the server.
- Specific behavior belongs to individual locations.

This avoids repeating the same directives and allows specific rules to override general ones.

Why are locations inside server blocks?

Because a location represents only a part of a website. It inherits the server's general configuration and overrides only the settings that need to change.

Why do location directives override server directives?

The server provides the default behavior for the entire website, while a location represents a more specific case. In configuration systems, the more specific rule takes precedence over the more general one.

What does "serving multiple websites" mean?

A single web server process can host multiple independent websites (virtual hosts). It chooses the appropriate server block for each request based on the listening port and the `Host` header.

What is the relationship between a server block and a website?

In webserv (and Nginx), one server block typically represents the configuration of one website (or virtual host). Multiple server blocks allow one executable to serve multiple websites.

What is the difference between a server block and a location block?

- **Server block:** Describes the entire website.
- **Location block:** Describes how a specific URL path within that website should behave.

What is the difference between a directive and a block?

- **Directive:** A single configuration instruction.
- **Block:** A container that groups multiple related directives together.

What is the difference between a URL path and a filesystem path?

- **URL path:** The path requested by the client (e.g., `/images/logo.png`).
- **Filesystem path:** The actual location of the resource on the server (e.g., `/var/www/images/logo.png`).

The server maps the URL path to the appropriate filesystem path according to its configuration.

How does a request use the configuration?

When a request arrives, the server: 1. Chooses the matching server block. 2. Chooses the best matching location block. 3. Starts with the server configuration. 4. Applies any overrides from the location. 5. Processes the request using the resulting configuration.